Degree-of-Polarization-Based Radio Frequency Interference Detection for Synthetic Aperture Radar

Yu Hashimoto, *Graduate Student Member, IEEE*, Akira Hirose, *Fellow, IEEE*, and Ryo Natsuaki, *Senior Member, IEEE*

***Abstract*— In microwave remote sensing, radio frequency interference (RFI) caused by other microwave systems has been a critical issue. RFI degrades the quality of synthetic aperture radar (SAR) images and disturbs phase and polarimetric information. Therefore, RFI must be detected and mitigated correctly to analyze SAR data accurately. In this article, we propose a novel RFI detection method based on the polarization states of received signals in either the time or frequency domain. Our study demonstrates that RFI from artificial sources is often more polarized than radar echoes from natural scatterers. We use the degree of polarization (DoP) to measure the polarization states and detect highly polarized RFI. The simulation results indicate that the proposed method was more accurate than the conventional power-spectrum-based methods and improved the detection limit by 4 dB compared to the conventional method based on the similarity between polarizations. We demonstrate the effectiveness of the proposed method by using data obtained by Phased Array type L-band SAR (PALSAR) aboard Advanced Land Observing Satellite (ALOS).**

***Index Terms*— Degree of polarization (DoP), polarimetric synthetic aperture radar (PolSAR), radio frequency interference (RFI), synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) has been widely used in various applications such as disaster monitoring, environmental monitoring, and ground mapping with its ability to observe the Earth's surface day and night, regardless of weather conditions. Polarimetric SAR (PolSAR) is an SAR system that transmits and receives signals with multiple polarizations, and it can extract additional information on the scattering properties of the observed area, which cannot be obtained by single-polarization SAR. However, the frequency band used by SAR systems is not exclusive to them. The band is shared with other systems such as global navigation satellite systems (GNSS), telecommunication, and broadcasting systems. Therefore, radio frequency interference (RFI) has been a critical issue for SAR systems [2]. Incoherent and weak RFI can be naturally suppressed down to the noise level through the 2-D coherent SAR focusing process. However, RFI is often strong enough to affect SAR images because radar echoes are weak in power. As an example, Fig. 1 shows a Pauli decomposition image of Barrow, Alaska observed by Phased Array type L-band SAR (PALSAR) aboard Advanced Land Observing Satellite (ALOS) in 2009. RFI generates unnatural patterns in the image and appears as bright linear or haze-like artifacts. The adverse effect of RFI on SAR systems is not limited to the image quality. It also disturbs the phase and polarimetric information of the radar echoes, which is essential for interferometric SAR (InSAR) and PolSAR applications [3], [4]. Moreover, strong RFI may cause saturation of receivers, which results in the loss of radar echoes. The situation of frequency sharing and the problem of RFI will become more severe with the increasing use of microwave systems. Therefore, accurate detection and mitigation of RFI are essential for SAR systems.

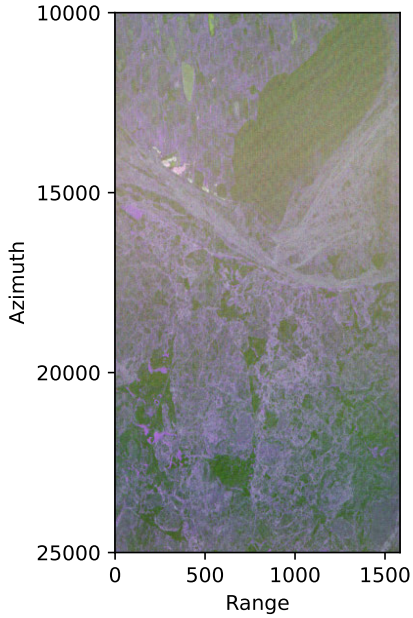


Fig. 1. Pauli decomposition image of a PALSAR scene in Barrow, Alaska (ALPSRP172601440). The area within 10 000–25 000th azimuth lines is shown.

Various methods have been proposed for RFI detection and mitigation in SAR data. These methods can be classified into

Received 19 September 2024; revised 3 April 2025; accepted 13 May 2025. Date of publication 15 May 2025; date of current version 29 May 2025. This work was supported by JSPS KAKENHI under Grant 23H00487. The ALOS original data are copyrighted by the Japan Aerospace Exploration Agency (JAXA) and METI, and provided under JAXA EORA4 Research Announcement PI No. ER4A2N006. An earlier version of this paper was presented in part at the 2024 IEEE International Geoscience and Remote Sensing Symposium (IGARSS 2024) [1]. *(Corresponding author: Yu Hashimoto.)*

The authors are with the Department of Electrical Engineering and Information Systems, The University of Tokyo, Tokyo 113-8556, Japan (e-mail: yu-hashimoto@eis.t.u-tokyo.ac.jp; ahirose@ee.t.u-tokyo.ac.jp; natsuaki@ee.t.u-tokyo.ac.jp).

Digital Object Identifier 10.1109/TGRS.2025.3570493

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:01:14 UTC from IEEE Xplore. Restrictions apply.

three categories: parametric, semiparametric, and nonparametric methods. Parametric methods use mathematical models to estimate the waveforms of RFI and subtract them from received signals. In [5] and [6], RFI was modeled as the sum of sinusoidal waves. Iterative adaptive approach and orthogonal subspace projection were employed to mitigate RFI in [7]. Generally, parametric methods can achieve high accuracy in RFI detection by precise modeling, but they require high computational costs for parameter estimation. In addition, they are only suitable for narrowband interference (NBI). They are not effective against wideband interference (WBI) because the model complexity increases along with the bandwidth.

Semiparametric methods exploit the sparsity and low-rank properties of RFI and suppress them by solving optimization problems. Principal component analysis (PCA) was used to exploit the low-rank property of RFI signals [8]. Robust PCA was applied by using a radar echo dictionary based on the chirp scaling algorithm [9]. In [10] and [11], sparse recovery algorithms were used to detect RFI. Sparse iterative covariance-based estimation [12], alternating projection [13], and Bayesian learning [14] were applied to use the sparsity of RFI as well. These methods are accurate but still computationally expensive due to the prior dictionary learning process and iteration.

Nonparametric methods detect and mitigate RFI based on the actual characteristics of received signals without modeling. Many methods focus on the spectrum intensity of RFI. For example, in [15], the average power of the range spectrum in adjacent azimuth lines was calculated to detect the strong RFI signals. Similarly, the least mean square filter [16] and the median filter [17] were applied to detect abnormally strong RFI. Dividing the range spectrum into subbands was also effective for suppressing NBI [18]. In [19], based on the fact that the power of useful signals has Gaussian properties, the Z-test was performed to detect abnormally strong RFI. Kullback–Leibler divergence (KLD) was also used in the power spectrum domain to detect weak RFI more sensitively [20]. In [21], RFI was detected by focusing on the Doppler frequency of RFI and using autocorrelation in the azimuth frequency domain. It is also common to suppress RFI based on matrix decomposition such as eigenvalue decomposition [22], [23], complex empirical mode decomposition [24], and adaptive filtering [25]. Digital beamforming was also used to suppress RFI by steering the nulls to the directions of RFI [26], [27]. Nonparametric methods are suitable for WBI and also computationally efficient because they do not require parameter estimation or prior learning. However, they are usually based on the difference in power between RFI and radar echoes, so that they are not effective for RFI with comparable power to the radar echoes. Additionally, these methods are usually applied in the range frequency domain to distinguish NBI more effectively, and it has been difficult to detect RFI in real time. To overcome these limitations, we need to extract and use the intrinsic characteristics of RFI.

In PolSAR systems, cross-polarization channels are more vulnerable to RFI than co-polarization channels because radar echoes are weaker in cross-polarization channels. To detect RFI more sensitively and accurately, the fact that both co- and cross-polarization channels receive RFI simultaneously can be used [28]. Few studies have approached RFI detection by using multiple polarizations. Sidelobe cancellation was applied to PolSAR data to suppress RFI by regarding one polarization channel as the main channel and the other as the auxiliary channel [29]. Uozumi et al. [30] used the similarity between two polarization channels to detect RFI in both the time and frequency domains. Semiparametric methods such as PCA were applied to multiple polarization channels [31]. Although these methods use multiple polarization channels to detect and suppress RFI, they do not fully exploit the polarization states of RFI.

In this article, we propose a novel RFI detection method based on the polarization states of RFI in the raw domain before focusing. SAR receives the summation of millions of backscattered signals, which becomes relatively unpolarized at the receiver. However, RFI is often highly polarized because it is generated by artificial sources with specific polarizations such as linear or circular polarizations. Some studies outside the field of SAR use polarimetric properties to extract signals of interest. The polarization states of signals were utilized in spectrum sensing for cognitive radio [32], and this method was applied to RFI detection in radio interferometry for astronomy [33]. We bring this idea to the field of SAR and propose a method to detect RFI based on the degree of polarization (DoP) in either the range time or frequency domain [1]. The DoP is used to measure the polarization states of received signals and detect highly polarized RFI. In the frequency domain, the DoP of RFI-contaminated bands is significantly different from that of RFI-free bands, resulting in sensitive NBI detection. In the time domain, our method enables quick RFI detection by bypassing the Fourier transform.

In Section II, the proposed DoP-based RFI detection method is described in detail. Sections III and IV show the simulation and experimental results of the proposed method, respectively.

II. DoP-Based RFI Detection

In this section, we describe the proposed RFI detection method based on the DoP. The effect of RFI on the polarization states of received signals is analyzed using DoP in either the time or frequency domain. Fig. 2 shows the flowchart of the proposed method. Our method uses raw data to extract the characteristics of received signals.

A. Time Domain Processing

Radar echoes are the sum of the backscatterings from numerous ground scatterers within the wide illuminated area of SAR, and they are relatively unpolarized. However, the artificial sources of RFI transmit polarized signals. Therefore, the polarization states of received signals contaminated by RFI are different from those of the RFI-free radar echoes.

PolSAR receives radar echoes with multiple polarizations, so the polarization states of the received signals can be analyzed. The received signal E^r can be modeled as follows:

$$E^r = S + N + I \tag{1}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:01:14 UTC from IEEE Xplore. Restrictions apply.

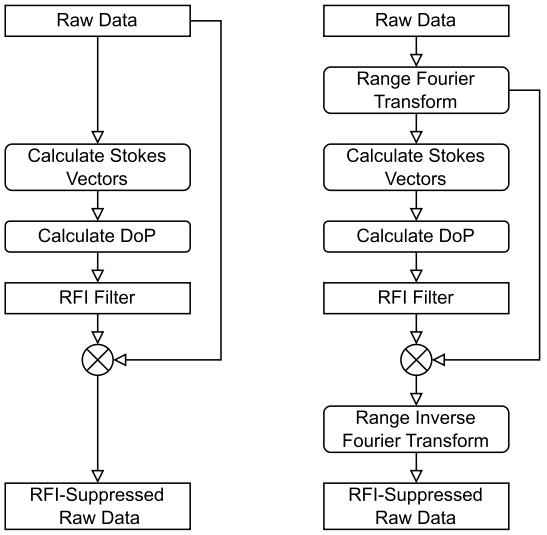


Fig. 2. Flowcharts of the DoP-based RFI detection method in (a) time and (b) frequency domains.

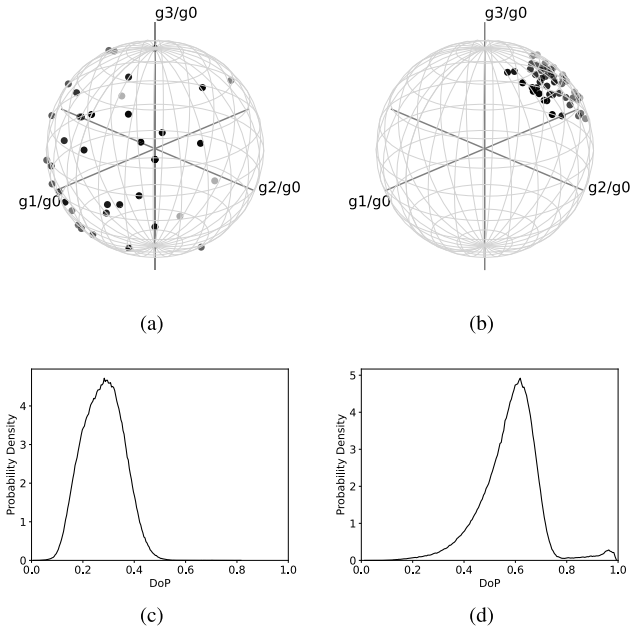


Fig. 3. (a) and (b) Normalized Stokes vectors of PALSAR data in the time domain plotted on Poincaré spheres. Signals (a) without RFI contamination and (b) with RFI. (c) and (d) Histograms of DoP in PALSAR scenes. Scene (c) without RFI (ALPSRP041577130) and (d) with RFI (ALPSRP172601440).

where S is the radar echo, N is the noise, and I is the RFI. If we assume a PolSAR system with horizontal (H) and vertical (V) polarization channels, the Jones vector $\boldsymbol{E^r}$ is expressed as follows:

$$\boldsymbol{E^r} = \begin{bmatrix} E^r_{\mathrm{H}} \\ E^r_{\mathrm{V}} \end{bmatrix} \tag{2}$$

where E^r_{H} and E^r_{V} are the H and V components of the received signal, respectively. The Stokes vector $\boldsymbol{g}$ is defined as follows:

$$\boldsymbol{g} = \begin{bmatrix} g_0 \\ g_1 \\ g_2 \\ g_3 \end{bmatrix} = \begin{bmatrix} |E^r_{\mathrm{H}}|^2 + |E^r_{\mathrm{V}}|^2 \\ |E^r_{\mathrm{H}}|^2 - |E^r_{\mathrm{V}}|^2 \\ 2\mathrm{Re}\{E^r_{\mathrm{V}} E^{r*}_{\mathrm{H}}\} \\ 2\mathrm{Im}\{E^r_{\mathrm{V}} E^{r*}_{\mathrm{H}}\} \end{bmatrix} \tag{3}$$

where * denotes the complex conjugate and Re and Im denote the real and imaginary parts, respectively. As an example, 50 normalized Stokes vectors $[g_1/g_0, g_2/g_0,$ and $g_3/g_0]^\top$ without and with RFI contamination from the raw PALSAR data (ALPSRP172601440, HH and HV) are plotted on each Poincaré sphere in Fig. 3(a) and (b). In Fig. 3(a), Stokes vectors of the RFI-free signals are distributed widely on the sphere, indicating that the radar echoes are unpolarized. However, in Fig. 3(b), the Stokes vectors of the RFI-contaminated signals converge to a specific point on the sphere. This example demonstrates that, unlike the radar echoes, the received signals containing RFI are highly polarized.

The difference in the polarization states of the radar echoes and RFI can be measured by the DoP. The DoP is defined as follows:

$$\mathrm{DoP} = \frac{\sqrt{\langle g_1 \rangle^2 + \langle g_2 \rangle^2 + \langle g_3 \rangle^2}}{\langle g_0 \rangle} \tag{4}$$

where $\langle \cdot \rangle$ denotes the ensemble average. The DoP takes a value between 0 and 1, where 0 indicates completely unpolarized signals and 1 indicates completely polarized, coherent signals. Unpolarized radar echoes have a low DoP value, while highly polarized RFI has a high DoP value. The window for calculating the DoP is slid in the range direction pulse by pulse to exploit the temporal continuity of RFI. The temporal resolution for detecting RFI is determined by the window size. However, a smaller window contains fewer samples, which increases the DoP values of radar echoes and may cause false detection. Therefore, both the temporal resolution and the signal separation should be considered when determining the window size.

Fig. 3(c) shows the histogram of the DoP values in the entire RFI-free PALSAR scene (ALPSRP041577130, HH and HV). The DoP values ranged between 0.1 and 0.5. Fig. 3(d) shows the histogram of the DoP values in the entire RFI-contaminated PALSAR scene (ALPSRP172601440, HH and HV). In this case, there are two peaks in the histogram: one around 0.6 and the other close to 1.0. The latter peak is caused by the signals affected by RFI. These examples indicate that the DoP can be used to distinguish RFI from radar echoes in the time domain. We detect the RFI by setting a threshold on the DoP as described in Section II-C.

B. Frequency Domain Processing

The DoP-based RFI detection principle in the time domain can be extended to the range frequency domain. In the frequency domain, RFI appears in both H and V channel spectra. By calculating the Stokes vectors in each frequency bin, the effect of RFI on the polarization states can be analyzed. As an example, 50 normalized Stokes vectors without and with RFI contamination from the raw PALSAR data (ALPSRP172601440, HH and HV) in the frequency domain are plotted on each Poincaré sphere in Fig. 4(a) and (b). Again, in Fig. 4(a), the Stokes vectors of the RFI-free band

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:01:14 UTC from IEEE Xplore. Restrictions apply.

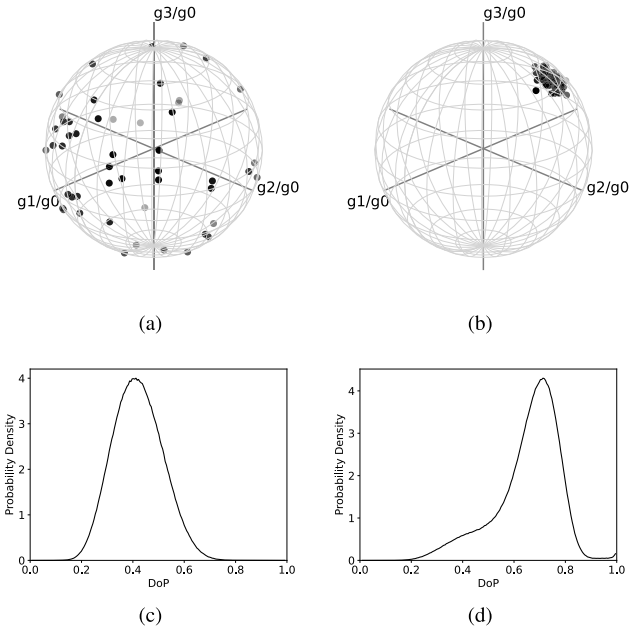


Fig. 4. (a) and (b) Normalized Stokes vectors of PALSAR data in the frequency domain plotted on Poincaré spheres. Signals (a) without RFI contamination and (b) with RFI. (c) and (d) Histograms of DoP in PALSAR scenes. Scene (c) without RFI (ALPSRP041577130) and (d) with RFI (ALPSRP172601440).

show inconsistent polarization states on the sphere. In contrast, in Fig. 4(b), the Stokes vectors of the RFI-contaminated band are concentrated on a specific point on the sphere, indicating that the RFI is highly polarized in the frequency domain as well as in the time domain.

The difference in the polarization states in the range frequency domain is also measured by the DoP. The window for calculating the DoP is set in the range frequency direction. Fig. 3(c) shows the histogram of the DoP values for the entire RFI-free PALSAR scene (ALPSRP041577130, HH and HV) in the frequency domain. The DoP values are distributed between 0.2 and 0.7. Fig. 3(d) shows the histogram of the DoP values for the entire RFI-contaminated PALSAR scene (ALPSRP172601440, HH and HV) in the frequency domain. Again, two peaks are observed in this histogram: one around 0.7 and the other close to 1.0, and the second peak is from the RFI-contaminated signals. Therefore, RFI can be detected by a threshold on the DoP in the frequency domain as well.

The time and frequency domain detection exhibits different characteristics depending on the bandwidth and duration of RFI. WBI with a short duration is easier to detect in the time domain, while NBI is easier to detect in the frequency domain. That is, NBI often has a high signal-to-interference ratio (SIR) in the time domain, but its SIR in the frequency domain is lower because the bandwidth of NBI is narrower than that of the radar echoes. Therefore, NBI can be detected with high sensitivity in the frequency domain.

C. Adaptive Threshold on DoP

The DoP-based RFI detection method requires a threshold on the DoP to distinguish RFI from radar echoes. It is important to determine an appropriate threshold to minimize both the false detection of the radar echoes and the overlooking of the RFI. In addition, the DoP values are also affected by the polarimetric properties of the radar echoes. The suitable threshold may vary depending on the characteristics of the observed scene and the ground surface. A fixed threshold reduces computational costs and is easy to implement, but it may not be effective in all situations. For instance, homogeneous scatterers in a wide area, such as a sea surface, can result in high DoP values, making it difficult to detect RFI with a fixed threshold.

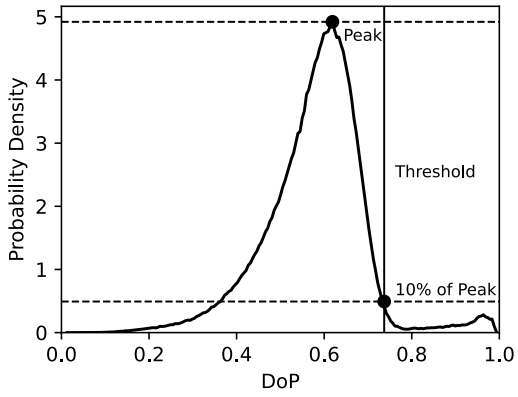


Fig. 5. Illustration of the adaptive threshold setting method based on the DoP distribution.

To address this issue, we propose an adaptive thresholding method based on the distribution of the DoP. A scene is divided into multiple regions in the azimuth direction, and the distribution of DoP values in each region is used to determine the threshold. Fig. 5 is an example of the adaptive threshold setting. The threshold is set at the point where the DoP distribution of radar echoes decreases to a certain level from the peak, for example, 10% in this article. The bold line represents the DoP distribution of received signals, and the two dashed lines are the peak and 10% of the peak.

With smaller azimuth blocks, the threshold can be set more adaptively according to the local characteristics of the scene. However, fewer samples in a smaller block may yield unreliable thresholds. The block size should be determined considering this trade-off. This adaptive thresholding method can prevent excessive false detection of radar echoes and improve the detection performance of the RFI. The percentage for setting the threshold should be determined carefully as well. A smaller percentage reduces unavoidable false detection of radar echoes but may overlook RFI signals with relatively low DoP values, which are mostly weak RFI. Also, the percentage should be higher than the local minimum between the peaks for this method to work correctly.

III. Simulation Experiments

To evaluate our proposed method, we conducted a simulation of RFI contamination using the PALSAR data without RFI (ALPSRP041577130). The PALSAR specifications are shown in Table I. The center frequency is 1270 MHz (L-band), and the bandwidth is 14 MHz. The range sampling rate is 16 MHz. Table II shows the details of three scenes used in the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:01:14 UTC from IEEE Xplore. Restrictions apply.

TABLE I
PALSAR SPECIFICATIONS

Center Frequency	1270 MHz (L-Band)
Bandwidth	14 MHz
Range Sampling Rate	16 MHz
Polarization channels	HH, HV, VH, VV

TABLE II
SCENE SPECIFICATIONS

Processing Level	Level 1.0
Scene ID	ALPSRP041577130
Observed Place	Manaus, Brazil
Observation Date	2006-11-5
Azimuth × Range	31538 × 1584 pixels
Features	Almost no RFI.
Scene ID	ALPSRP172601440
Observed Place	Barrow, Alaska, United States
Observation Date	2009-4-21
Azimuth × Range	31417 × 1584 pixels
Features	Affected by RFI.
Scene ID	ALPSRP071951440
Observed Place	Barrow, Alaska, United States
Observation Date	2007-6-1
Azimuth × Range	31418 × 1584 pixels
Features	Affected by RFI.
	A large sea area with consistent scattering.

TABLE III
PARAMETERS FOR SIMULATED RFI SIGNALS

NBI (FM chirp)	Bandwidth	1 MHz (5–6 MHz)
	Chirp rate	2.1×10^{10} Hz/s
WBI (OFDM)	Bandwidth	6.4 MHz (1–7.4 MHz)
	Subcarriers	64
Azimuth		20000th line
Range		250–1000 samples
Duration		750 px (46.9 µs)
Polarization		LHC, H

following simulation and experiments. All the scenes used in this article are the Level 1.0 products and full polarimetric. The scene ALPSRP041577130 was observed in Manaus, Brazil, and almost no RFI was recorded, which makes it suitable for the simulation.

We superimposed simulated RFI signals on HH and HV channels of the PALSAR data by shifting SIR. Then, our DoP-based method and the conventional Z-test-based [19], similarity-based [30], and KLD-based [20] methods were applied to the contaminated data. The simulation was conducted on four different conditions. We used two RFI types with two different polarizations. That is, the RFI types were frequency-modulated (FM) chirp as NBI and orthogonal frequency division multiplexing (OFDM) as WBI whose polarization was either left-handed circular (LHC) or horizontal (H). The parameters for the simulated RFI signals are shown in Table III. The NBI signal has a chirp rate of 2.1×10^{10} Hz/s and a bandwidth of 1 MHz. The WBI signal has a bandwidth of 6.4 MHz and 64 subcarriers. Both RFI signals were added to the 20 000th azimuth line between 250 and 1000 range samples. The simulated RFI signals $\boldsymbol{I}$ are expressed as follows:

$$\boldsymbol{I} = \begin{bmatrix} I_{\mathrm{H}} \\ I_{\mathrm{V}} \end{bmatrix} = A\ f(\tau)\ \boldsymbol{p} \tag{5}$$

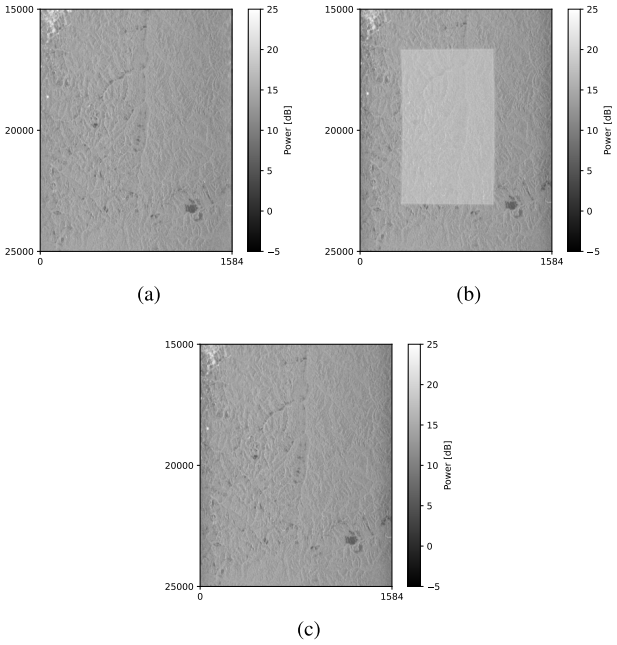


Fig. 6. SAR images with simulated FM chirp RFI signals (HH channel). (a) Before adding RFI, and after adding RFI (b) at the maximum power of SIR = −40 dB and (c) at the detection limit of SIR = 5 dB.

where A is the amplitude, $f(\tau)$ is the RFI signal, and $\boldsymbol{p}$ is the polarization vector. The polarization vector $\boldsymbol{p}$ is $[1/\sqrt{2},\ j/\sqrt{2}]^{\top}$ for LHC and $[1,\ 0]^{\top}$ for H pol. The amplitude of RFI signals ranged from −40 to 20 dB as in the time-domain SIR. The SIR in the time domain is defined as follows:

$$\mathrm{SIR} = 10 \log_{10} \left(\frac{\bar{P}_S}{A^2} \right) \tag{6}$$

where $\bar{P}_S$ is the average power of the radar echoes in the time domain. Fig. 6 shows examples of the HH channel SAR images without and with the generated NBI signals. Fig. 6(b) shows the original image before adding RFI. Fig. 6(c) shows the image with RFI at SIR = −40 dB, and there is a large white area due to the simulated RFI. Fig. 6(d) shows the image with RFI at SIR = 5 dB, and the RFI is too weak to be visually confirmed.

Considering the trade-off between the detection resolution and the signal separation, we experimentally set the DoP window sizes to 50 pixels and 20 pixels for the time and frequency domains, respectively. Thresholds of the DoP were determined experimentally as well. For the fixed thresholds, they were set to 0.6 and 0.8 for the time and frequency domains, respectively. For the adaptive thresholds, the DoP distribution was calculated for each block of 1000 azimuth lines, and the threshold was set to the point where the probability density of DoP decreased to 10% from the peak.

A. Evaluation Method for Simulation Results

The detection results of the simulated RFI were evaluated using binary classification metrics because the temporal and spectral properties of the RFI signals were known. For the time domain, the truth for the binary classification was defined as the pixels where the RFI was added. For the frequency

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:01:14 UTC from IEEE Xplore. Restrictions apply.

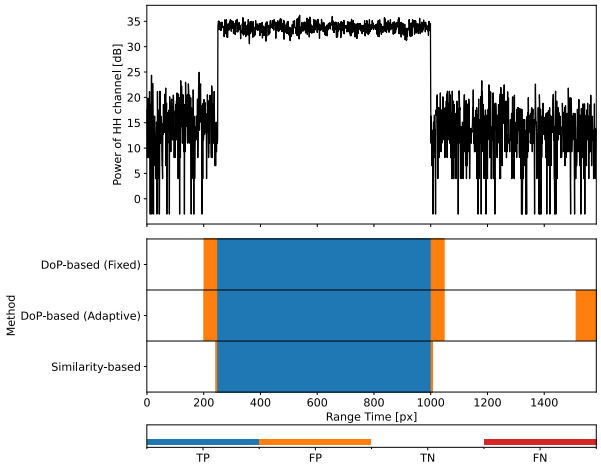


Fig. 7. Example of simulation results in the time domain (NBI, LHC, SIR $= -20$ dB). (Top) Intensity of the HH channel. (Bottom) Classification results of the DoP-based and similarity-based methods.

domain, the truth was defined as the frequency bins where the spectral power was increased by 6 dB or more. The binary classification metrics used in the evaluation were accuracy, Cohen's kappa, and F-score. The definitions of the metrics are expressed as follows:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \tag{7}$$

$$\kappa = \frac{2((\text{TP} \times \text{TN}) - (\text{FP} \times \text{FN}))}{(\text{TP} + \text{FN}) \times (\text{FN} + \text{TN}) + (\text{TP} + \text{FP}) \times (\text{FP} + \text{TN})} \tag{8}$$

$$F = \frac{\text{TP}}{\text{TP} + \frac{1}{2}(\text{FP} + \text{FN})} \tag{9}$$

where TP, TN, FP, and FN are the number of true positives, true negatives, false positives, and false negatives, respectively. The accuracy is the ratio of the correct classification to the total number of samples. In this simulation, the SIR is shifted from -40 to 20 dB, so the ratio of the two classes is unbalanced in the frequency domain due to the sidelobes of the RFI. Therefore, we employed Cohen's kappa and F-score as evaluation metrics to consider the unbalanced classes. Cohen's kappa is a measure of the agreement between two classifiers. It takes values between -1 and 1, and higher values indicate better agreement. The F-score is the harmonic mean of precision and recall, and it takes values between 0 and 1. It indicates higher values when FP and FN are small.

B. Simulation Results

Fig. 7 shows an example of the detection results for the LHC pol. NBI in the time domain at SIR $= -20$ dB. The top panel shows the raw intensity of the HH channel, and the signal power increased during the existence of the superimposed RFI between 250 and 1000 samples. The bottom panel shows the classification results of each method. Blue, orange, and red represent TP, FP, and FN, respectively, while TN is left blank. In this example, the conventional similarity-based method detected the RFI accurately, while our DoP-based methods had some false detections due to the large window size. The DoP-based method with adaptive threshold falsely classified samples near the end of the line as RFI. This is because the adaptive threshold method sets the value at 10% of peak, which causes a small number of false detections by design. The DoP values of those samples were higher than the rest of the radar echoes, and they were falsely detected as RFI. Fig. 8 shows the evaluation results of the time domain. Fig. 8(a) and (b) shows the evaluation results for NBI with LHC and H polarizations, respectively. Fig. 8(c) and (d) shows the evaluation results for WBI with LHC and H polarizations, respectively. We compared accuracy, Cohen's kappa, and F-score while shifting the SIR. For LHC pol., the conventional similarity-based method achieved the highest accuracy against strong RFI from -40 to -20 dB, but both the DoP-based methods with fixed and adaptive threshold also achieved comparable accuracy around 0.9. Although the similarity-based method could not detect RFI with H pol., the DoP-based methods were able to detect it with comparable sensitivity to LHC pol. The adaptive DoP-based method was the most effective against weak RFI below -5 dB. The results indicate that our DoP-based methods are effective in detecting RFI in the time domain, and the adaptive threshold improves the sensitivity against weak RFI.

Fig. 9 shows an example of the detection results for the LHC pol. NBI in the frequency domain at SIR $= -20$ dB. The top panel shows the power spectrum of the HH channel, and the band from 5 to 6 MHz was contaminated by RFI. The bottom panel shows the classification results of each method. In this example, the proposed DoP-based methods accurately detected the RFI. In particular, the DoP-based method with adaptive threshold was able to detect the sidelobes of the RFI around -8 and 8 MHz, which were not abnormal in the power spectrum. The conventional Z-test-based and KLD-based methods were too sensitive and caused false positives around the main lobe. Fig. 10 shows the evaluation results of the frequency domain. Note that the SIR values are expressed in the time domain so that the results can be compared between the time and frequency domains. In Fig. 10(a) and (b), the DoP-based methods and the conventional similarity-based and Z-test-based methods had high accuracy, but it was due to the unbalanced classes of the NBI. That is, any method that overlooks the RFI will have high accuracy up to 0.9 because the band of RFI only occupies about 10% of the entire frequency band. Cohen's kappa and F-score should be considered for the evaluation. For NBI, Fig. 10(a) and (b) shows that the proposed DoP-based methods achieved better F-score than the three conventional methods in the wide SIR range from -40 to 4 dB. Our DoP-based method with adaptive threshold detected weak RFI at 5 dB, which was not visible in the SAR image in Fig. 6(c). The adaptive threshold significantly improved the detection performance against weak RFI from 1 to 4 dB. In the case of WBI, the results were not as good as NBI overall. The adaptive DoP-based method had the highest F-score in the wide SIR range, but the values were low and insignificant.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:01:14 UTC from IEEE Xplore. Restrictions apply.

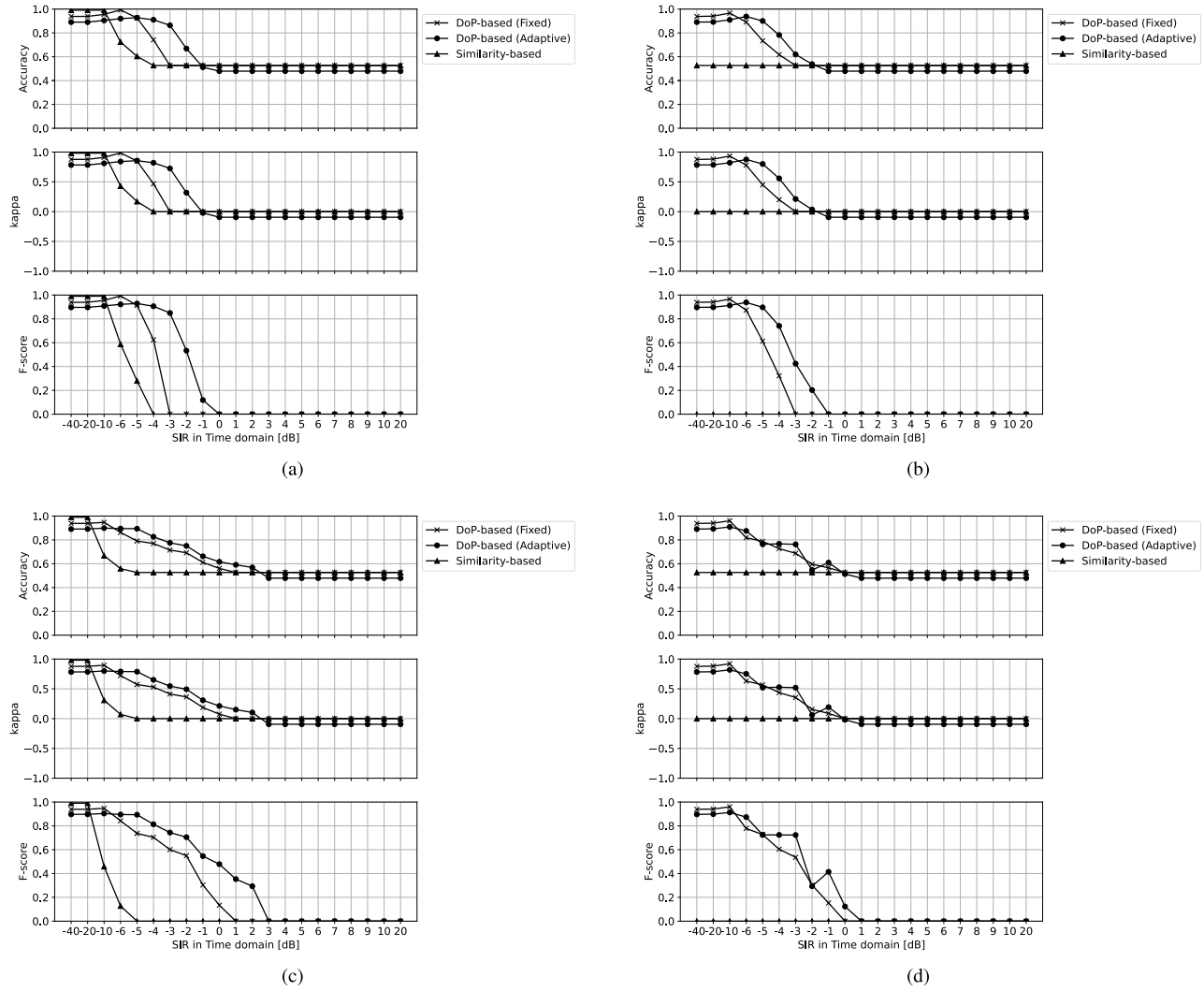


Fig. 8. Evaluation of simulation results in the time domain. (a) and (b) NBI with LHC and H pol. (c) and (d) WBI with LHC and H pol.

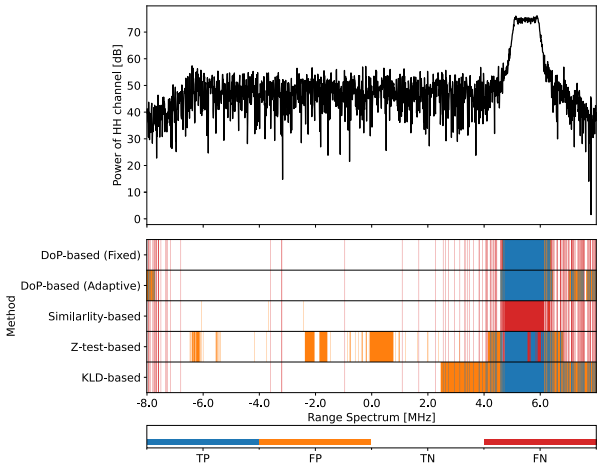


Fig. 9. Example of simulation results in the frequency domain (NBI, LHC, SIR $= -20$ dB). (Top) Power spectrum of the HH channel. (Bottom) Classification results of the DoP-based, Z-test-based, similarity-based, and KLD-based methods.

Our DoP-based methods were able to detect strong WBI from -40 to -10 dB, but they failed to detect the WBI as the SIR increased. The conventional Z-test-based and KLD-based methods, however, had false positives outside the WBI band in all SIR ranges, which resulted in low F-scores. The similarity-based method was not consistent in detection, and both the true and false positives were few in all SIR ranges.

By comparing the results of the time and frequency domains, time domain detection is effective against strong NBI with SIR ≤ 0 dB, and frequency domain detection is effective against weak NBI with SIR ≥ 0 dB. The results indicate that the time domain detection is suitable for WBI. Our DoP-based methods were able to detect RFI in both domains more effectively than the conventional methods, and applying the adaptive threshold improved the detection performance against weak RFI.

IV. Experiment With PALSAR Data Containing RFI

We conducted experiments using the PALSAR scenes with real RFI contamination (ALPSRP172601440, ALPSRP071951440). As shown in Table II, the scenes ALPSRP172601440 and ALPSRP071951440 were observed in Barrow, Alaska, and RFI was recorded in both scenes.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:01:14 UTC from IEEE Xplore. Restrictions apply.

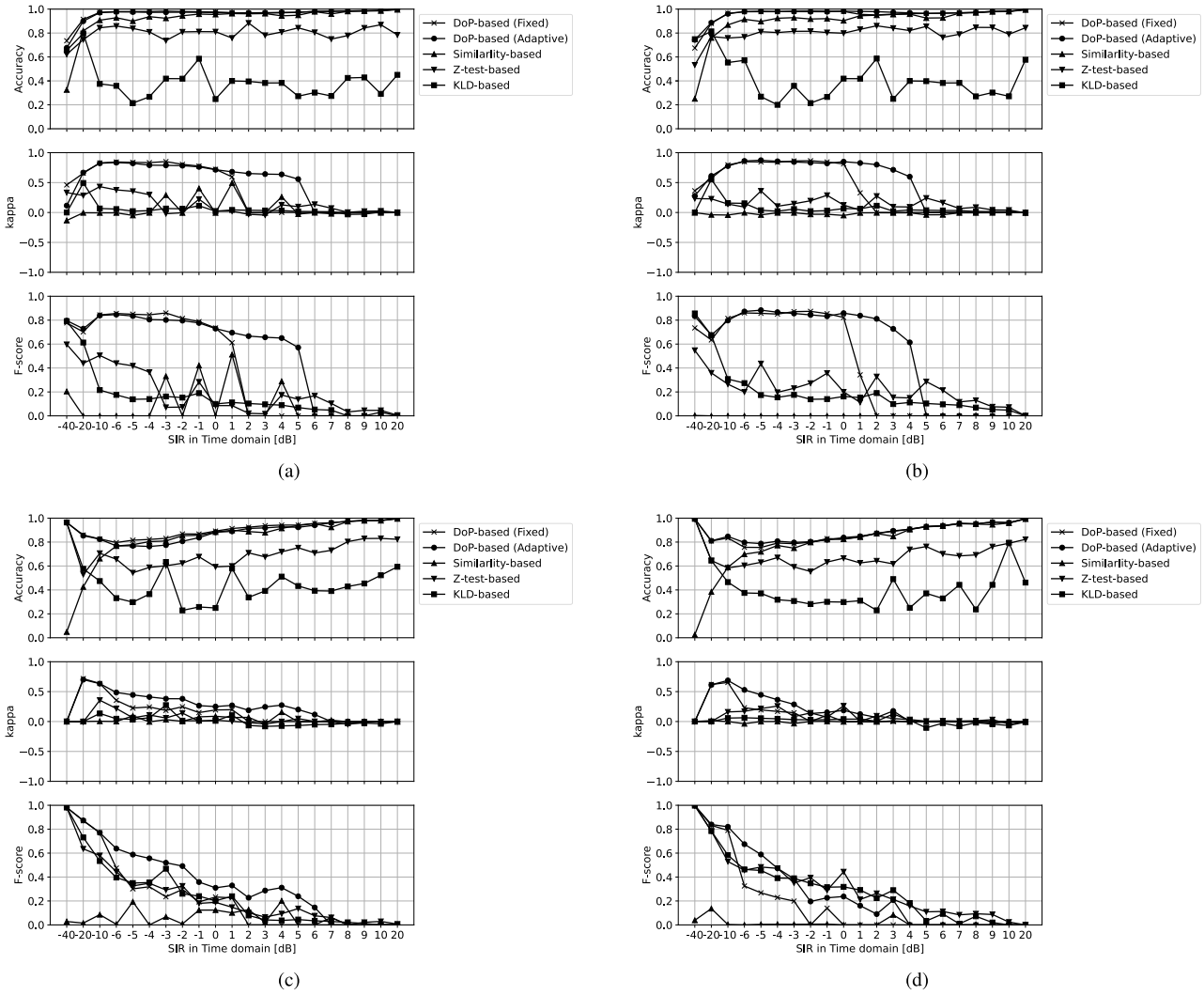


Fig. 10. Evaluation of simulation results in the frequency domain. (a) and (b) NBI with LHC and H pol. (c) and (d) WBI with LHC and H pol.

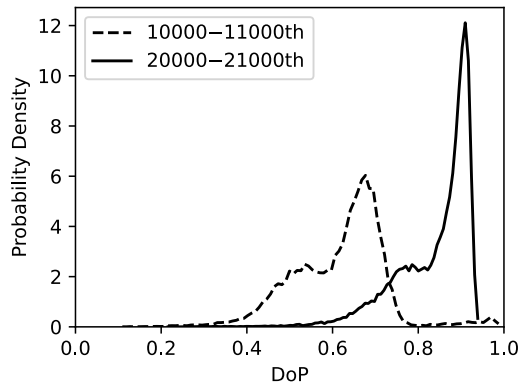


Fig. 11. Histograms of DoP within two different azimuth areas in the PALSAR scene ALPSRP071951440. The sea area from 20 000th to 21 000th azimuth lines has higher DoP values than the land area from 10 000th to 11 000th azimuth lines.

ALPSRP172601440 was used in the experiments of the previous studies [19], [30]. In ALPSRP071951440, a large area with consistent scatterings was observed. As shown in Fig. 11, the DoP values of the radar echoes from a part of the sea area (20 000–21 000th azimuth lines) were significantly higher than those from other areas. This scene was used to prove the feasibility of the proposed method. Our DoP-based method and the conventional Z-test-based [19], similarity-based [30], and KLD-based [20] methods were applied to the contaminated data to evaluate the mitigation performance.

The parameters for the DoP-based method were determined experimentally as in the simulation. The window size for calculating DoP was set to 100 pixels in the time domain and 20 pixels in the frequency domain. The thresholds for the DoP were set to 0.75 and 0.9 for the time and frequency domains, respectively. The adaptive thresholds were set as described in Section II-C. Distributions of DoP were calculated for each block of 1000 azimuth lines, and the threshold was set to the point where the probability density of DoP decreased to 10% from the peak.

A. Evaluation of RFI Mitigation Using Interferometric Coherence

To evaluate the performance of RFI mitigation, the interferometric coherence of cross-polarization (HV, VH) single-look

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:01:14 UTC from IEEE Xplore. Restrictions apply.

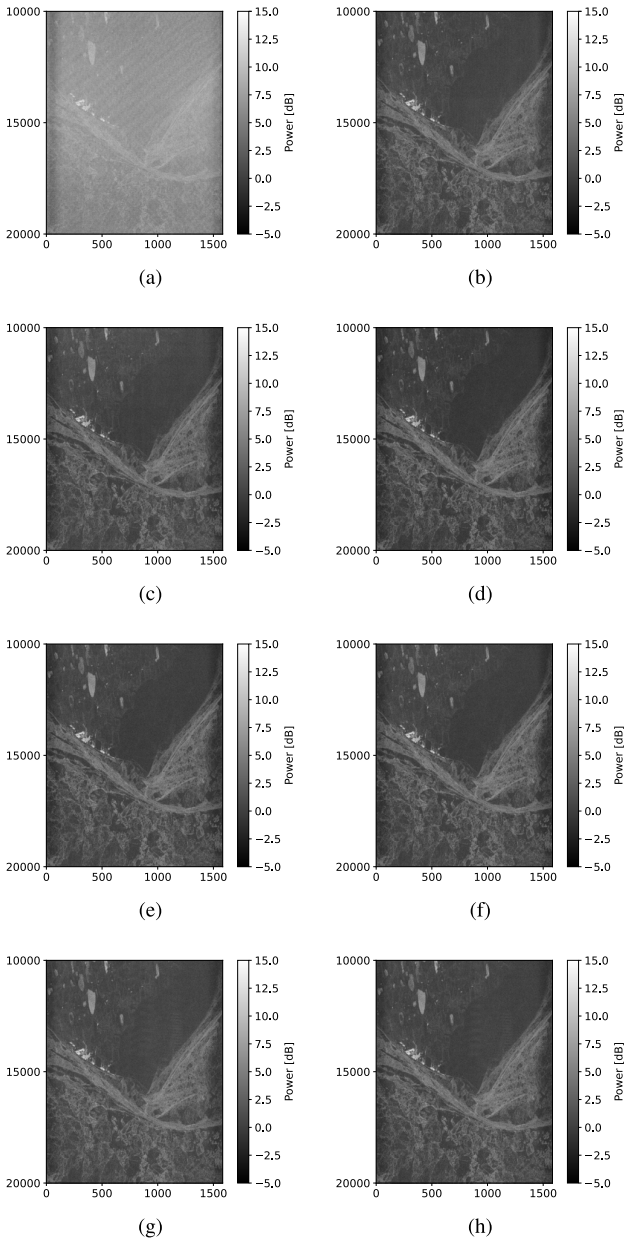


Fig. 12. Comparison of SAR images (HV channel) for ALPSRP172601440. (a) Before RFI mitigation. After RFI mitigation with (b) Z-test-based, (c) similarity-based, (d) KLD-based, time-domain DoP-based method with (e) fixed and (f) adaptive threshold, and frequency-domain DoP-based method with (g) fixed and (h) adaptive threshold method.

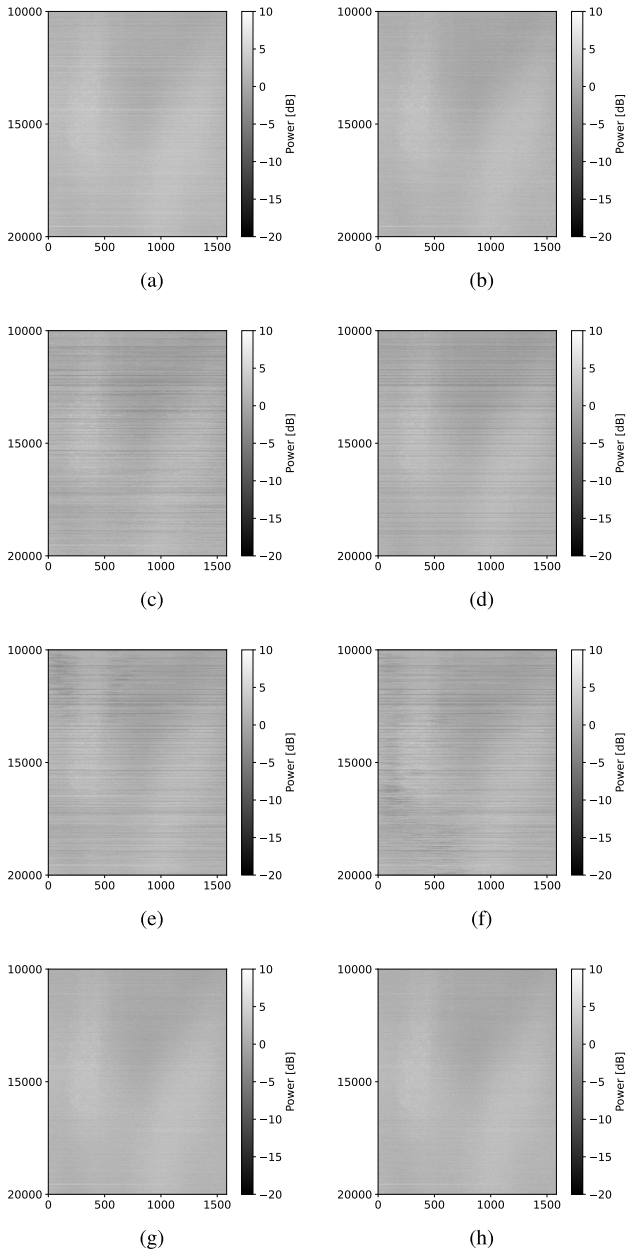


Fig. 13. Comparison of raw intensity in the time domain (HV channel) for ALPSRP172601440. (a) Before RFI mitigation. After RFI mitigation with (b) Z-test-based, (c) similarity-based, (d) KLD-based, time-domain DoP-based method with (e) fixed and (f) adaptive threshold, and frequency-domain DoP-based method with (g) fixed and (h) adaptive threshold method.

complex (SLC) images can be used [30]. The interferometric coherence γ of the cross-polarization SLC data $X_{\mathrm{HV}}, X_{\mathrm{VH}}$ is expressed as follows:

$$\gamma = \frac{\left|\sum_{i=1}^{W} X_{\mathrm{HV}}(i)\ X_{\mathrm{VH}}^{*}(i)\right|}{\sqrt{\sum_{i=1}^{W} |X_{\mathrm{HV}}(i)|^2}\sqrt{\sum_{i=1}^{W} |X_{\mathrm{VH}}(i)|^2}} \tag{10}$$

where W is the number of pixels in the window, which was set to 5×5 pixels in this experiment. From the principle of the PolSAR theory, the scattering properties of a target against cross-polarizations are the same, resulting in high interferometric coherence values. However, RFI lowers the coherence values because it is time-varying and cross-polarization signals are received at different timings. Therefore, higher coherence values of cross-polarization SLC images indicate better RFI mitigation performance. We used histograms of the interferometric coherence values with 100 bins and their mean values to evaluate the RFI mitigation performance.

B. Experimental Results for ALPSRP172601440

Fig. 12 shows the focused SAR images of the HV channel before and after RFI mitigation. Figs. 13 and 14 show the raw intensity in the time domain and the raw power spectrum for the HV channel, respectively. (a) is the original image before RFI mitigation. (b)–(h) show the images after RFI mitigation

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:01:14 UTC from IEEE Xplore. Restrictions apply.

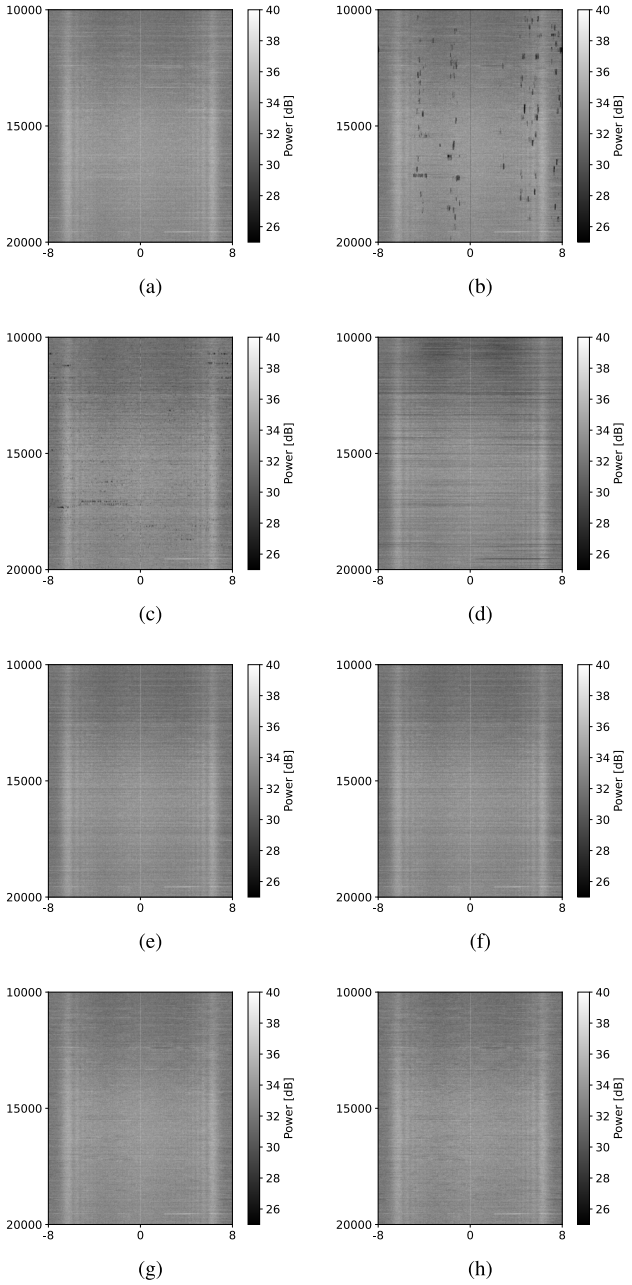


Fig. 14. Comparison of HV channel raw power spectra for ALPSRP172601440. (a) Before RFI mitigation. After RFI mitigation with (b) Z-test-based, (c) similarity-based, (d) KLD-based, time-domain DoP-based method with (e) fixed and (f) adaptive threshold, and frequency-domain DoP-based method with (g) fixed and (h) adaptive threshold method.

by the Z-test-based method, the similarity-based method, the KLD-based method, the time-domain DoP-based method with fixed and adaptive threshold, and the frequency-domain DoP-based method with fixed and adaptive threshold, respectively. The area within 10 000th–20 000th azimuth lines is shown in Figs. 12–14.

In Fig. 12(a), the effects of RFI are visible as diagonal stripes and haze-like artifacts. Each RFI detection method suppressed most of the RFI. In particular, our time-domain DoP-based method shown in Fig. 12(e) and (f) has good contrast around the lagoon area in the top right corner. However,

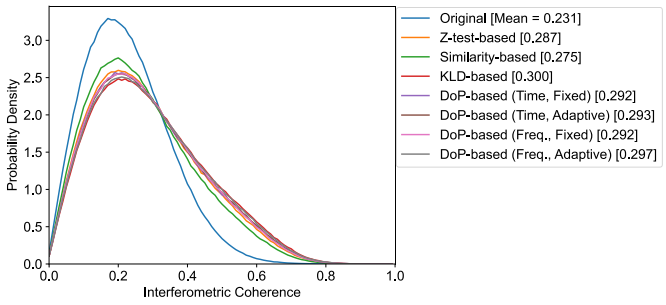


Fig. 15. Histograms and mean values of the interferometric coherence between cross-polarizations (HV, VH) SLC images for ALPSRP172601440.

as shown in Fig. 13(c)–(f), the conventional similarity-based and KLD-based methods and the time-domain DoP-based method caused entire signal loss at some azimuth lines. The signal loss is also visible in Fig. 14(c)–(f). The frequency-domain DoP-based method shown in Fig. 14(g) and (h), the conventional Z-test-based method in (b), and the KLD-based method in (d) have suppressed most of the RFI bands observed in the original image (a). Our frequency-domain DoP-based method did not suppress the dc component, which was mistakenly suppressed by the conventional Z-test-based method.

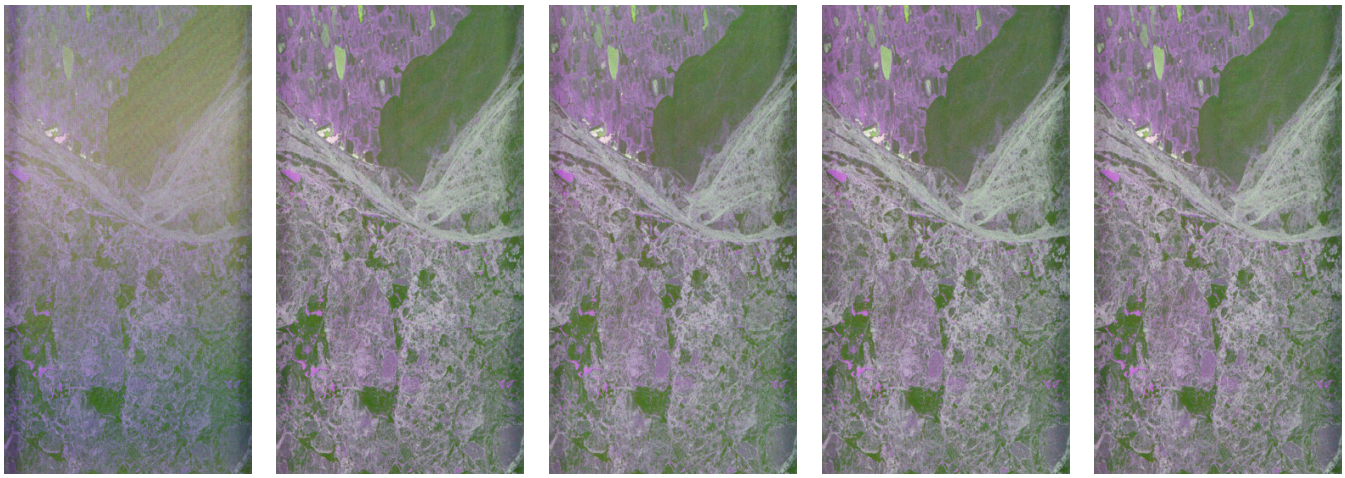
Fig. 15 shows the histograms of the interferometric coherence γ of the cross-polarization SLC images and their mean values. The original image has the lowest mean coherence value of 0.231 due to the RFI contamination. Each RFI mitigation method moved the histogram toward the right, and the mean coherence values increased from 0.231. Our DoP-based methods achieved higher values than the conventional similarity-based and Z-test-based methods. The DoP-based method in the frequency domain with adaptive threshold marked a comparable value to the highest of the conventional KLD-based method. The adaptive threshold improved the coherence results in both the time and frequency domains.

Fig. 16 shows Pauli decomposition images of (a) before RFI mitigation and after RFI mitigation with (b) Z-test-based, (c) similarity-based, (d) KLD-based, and (e) frequency-domain DoP-based method with adaptive threshold. Our DoP-based method and the three conventional methods suppressed most of the RFI artifacts that were visible in the original image. While the KLD-based method has no visible artifacts, the other methods have generated some artifacts in the lagoon area. Our DoP-based method generated horizontal stripes in the lagoon area, while the conventional methods generated vertical stripes. The horizontal stripes in the lagoon area in (e) were only visible in the frequency-domain DoP-based method, which were also visible in the SAR image in Fig. 12(g) and (h) but not in the time-domain DoP-based method (e) and (f). The patterns were probably caused by the residual of RFI pulses with short duration and wideband characteristics.

C. Experimental Results for ALPSRP071951440

Fig. 17 shows the focused SAR images of the HV channel before and after RFI mitigation in the case of ALPSRP071951440. Figs. 18 and 19 show the raw intensity in the time domain and the raw power spectrum for the HV

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:01:14 UTC from IEEE Xplore. Restrictions apply.



(a) (b) (c) (d) (e)

Fig. 16. Comparison of Pauli decomposition images for ALPSRP172601440. (a) Before RFI mitigation. After RFI mitigation with (b) Z-test-based, (c) similarity-based, (d) KLD-based, and (e) frequency-domain DoP-based method with adaptive threshold methods. The area within 10 000th–25 000th azimuth lines is shown in the figures.

channel, respectively. (a) is the original image before RFI mitigation. (b)–(h) shows the images after RFI mitigation by the Z-test-based method, the similarity-based method, the KLD-based method, the time-domain DoP-based method with fixed and adaptive threshold, and the frequency-domain DoP-based method with fixed and adaptive threshold, respectively. The area within 15 000th–25 000th azimuth lines is shown in Figs. 17–19.

In Fig. 17(a), a large gray area is visible between 15 000th and 22 000th lines in the center of the image. In a specific condition, the water surface induces the Bragg scattering and appears as if there are uniformly coherent scatterers. The RFI contamination is also visible as linear and haze-like artifacts. The DoP-based method with a fixed threshold in Fig. 17(e) and (g) excessively suppressed signals in the sea surface area because the DoP of the sea surface was high due to the Bragg scattering, which resulted in darker images. The signal loss is confirmed by the raw intensity images in Fig. 18(e) and the raw power spectra in Fig. 19(e) and (g). The adaptive threshold setting in Fig. 17(f) and (h) prevented excessive signal suppression. The frequency-domain DoP-based method with adaptive threshold in Fig. 17(h) has as good contrast as the conventional Z-test-based method in (b) and KLD-based method in (d). The similarity-based method in (c) has vertical linear patterns due to excessive suppression and residue of RFI in the bottom right corner.

Fig. 20 shows the actual threshold values of the adaptive DoP method in the time and frequency domains. The adaptive threshold method calculated the threshold for each 1000 azimuth lines. The threshold values around the sea area (15 000–25 000th azimuth lines) were adjusted to higher values than other areas. The radar echoes from the sea area show higher DoP values than other areas due to the Bragg scattering as shown in Fig. 11, and the adaptive method worked properly to prevent false detection of those echoes.

Fig. 21 shows the histograms of the interferometric coherence γ of the cross-polarization SLC images and their mean values. Our DoP-based method in both the time and frequency domains had higher mean coherence values than the conventional similarity-based method, with the best value of 0.316 for the adaptive frequency domain. While the DoP-based method in the time domain was less effective than the Z-test-based method, the adaptive frequency-domain DoP-based method ($\bar{\gamma} = 0.316$) was as effective as the Z-test-based method (0.315). The KLD-based method had the highest mean coherence value of 0.324.

TABLE IV
COMPUTATIONAL TIME

Method	Time
Z-test-based	1587 s
Similarity-based	1742 s
KLD-based	2808 s
DoP-based (Time, Adaptive)	1283 s
DoP-based (Frequency, Adaptive)	1122 s

Table IV summarizes the actual computation time of each method. Our DoP-based methods were faster than the three conventional methods, reducing the computation time by more than 50% compared to the KLD-based method. This is because our methods require fewer signal samples, simpler calculations, and even fewer filters. In addition, although the Z-test-based and KLD-based methods require a certain amount of samples as a population, our method does not require them if the threshold is predetermined. This is an advantage when detecting RFI in a short time series or pulse by pulse.

Fig. 22 shows Pauli decomposition images of (a) before RFI mitigation and after RFI mitigation with (b) Z-test-based, (c) similarity-based, (d) KLD-based, and (e) frequency-domain DoP-based method with adaptive threshold method. Our DoP-based method and the conventional Z-test-based method had similar results in suppressing RFI artifacts. The similarity-based method caused severe noise in the image due to the excessive signal suppression. For the KLD-based method shown in Fig. 22(d), the lagoon area in the top right corner had less green color, and the power balance between polarization

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:01:14 UTC from IEEE Xplore. Restrictions apply.

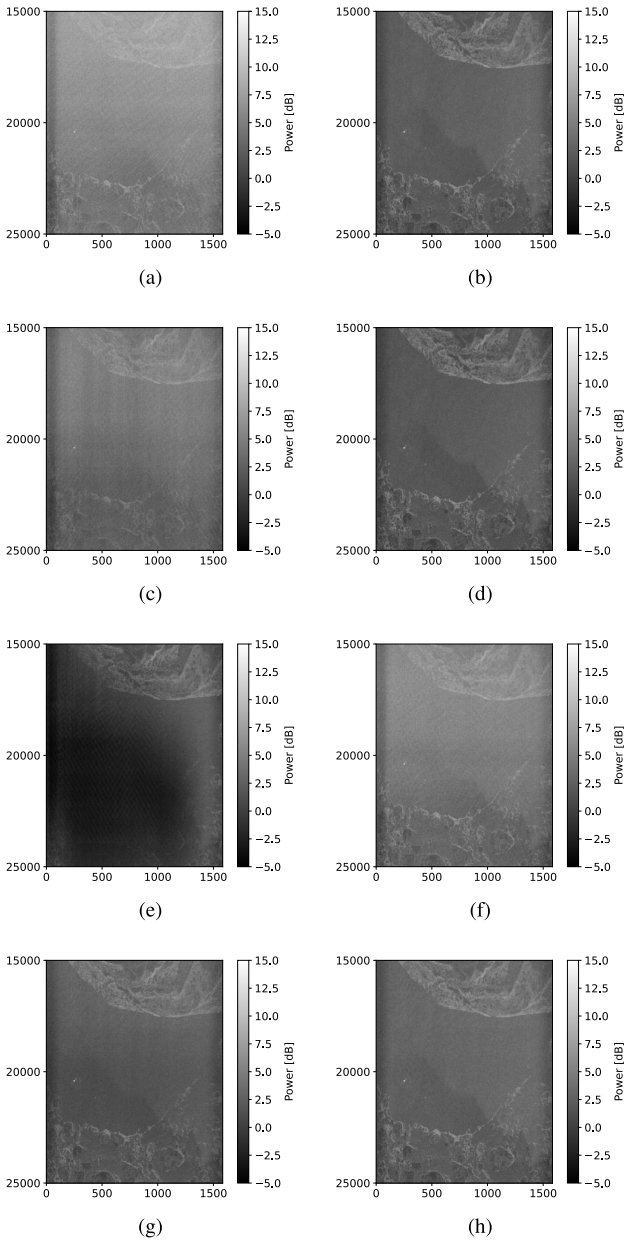


Fig. 17. Comparison of SAR images (HV channel) for ALPSRP071951440. (a) Before RFI mitigation. After RFI mitigation with (b) Z-test-based, (c) similarity-based, (d) KLD-based, time-domain DoP-based method with (e) fixed and (f) adaptive threshold, and frequency-domain DoP-based method with (g) fixed and (h) adaptive threshold method.

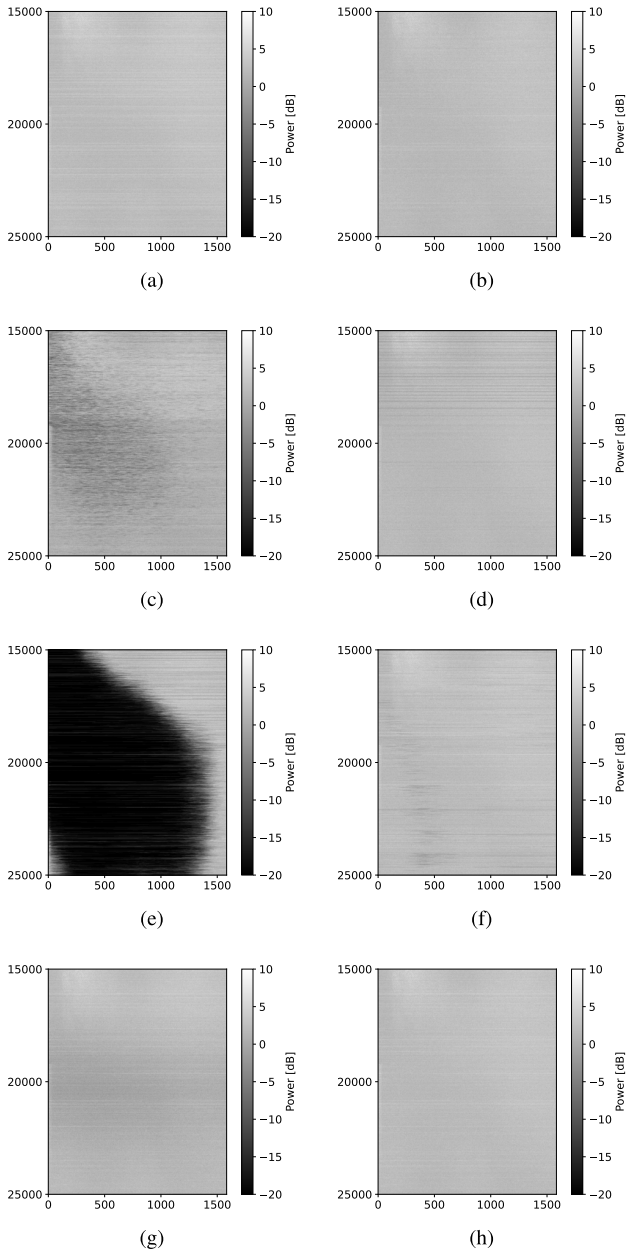


Fig. 18. Comparison of raw intensity in the time domain (HV channel) for ALPSRP071951440. (a) Before RFI mitigation. After RFI mitigation with (b) Z-test-based, (c) similarity-based, (d) KLD-based, time-domain DoP-based method with (e) fixed and (f) adaptive threshold, and frequency-domain DoP-based method with (g) fixed and (h) adaptive threshold method.

channels was not maintained. Similarly, the sea area in the lower half had less red and blue colors. This was caused by the excessive suppression in HH and VV channels. In short, the KLD-based method is probably too sensitive to the variation of the received power.

V. Discussion

A. Discussion on the Condition of the Simulation

In Section III, we simulated two types of RFI, NBI and WBI, with two different polarization states. However, actual RFI is diverse in bandwidth, duration, and polarization, and its impact varies depending on the situation of desired signals. For further evaluation with simulated RFI, it is necessary to characterize RFI in detail and conduct a simulation based on the study. In case the sniffing data containing real RFI is available, it will be effective to evaluate using the actual RFI data.

B. Discussion on the Experimental Results

In Section IV, we evaluated our DoP-based method by applying it to PALSAR data with real RFI contamination. Although both our method and the conventional similarity-based method use data from two polarizations, our method indicated better RFI detection performance in both

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:01:14 UTC from IEEE Xplore. Restrictions apply.

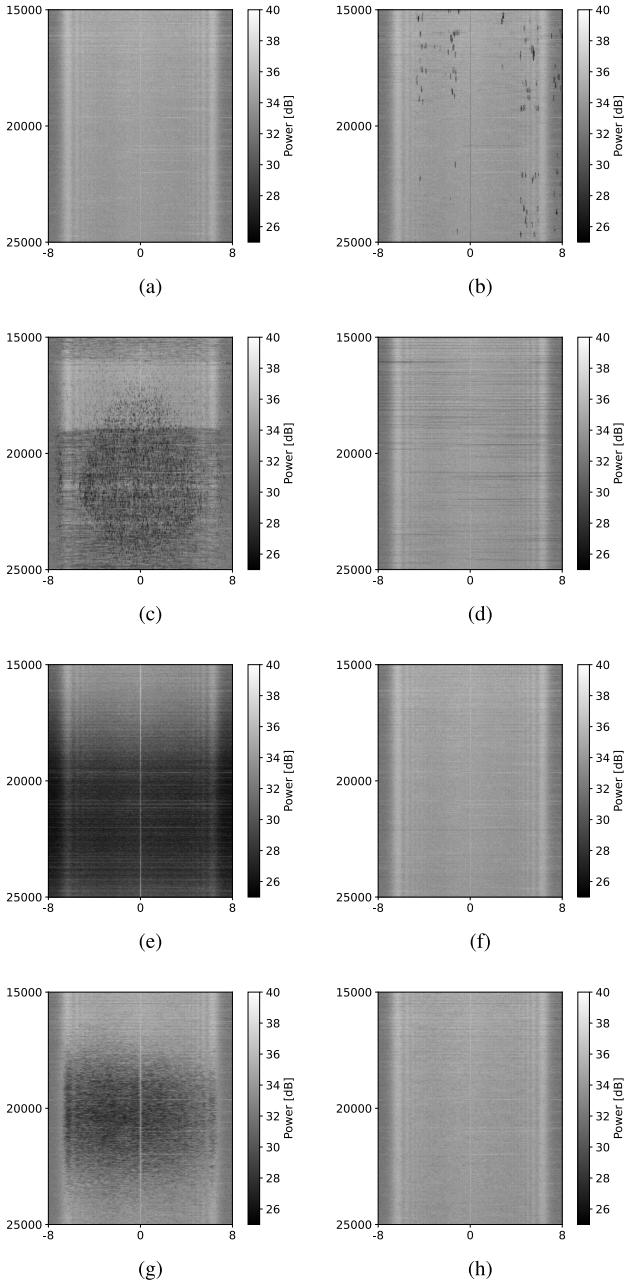


Fig. 19. Comparison of HV channel raw power spectra for ALPSRP071951440. (a) Before RFI mitigation. After RFI mitigation with (b) Z-test-based, (c) similarity-based, (d) KLD-based, time-domain DoP-based method with (e) fixed and (f) adaptive threshold, and frequency-domain DoP-based method with (g) fixed and (h) adaptive threshold method.

the time and frequency domain than the existing method. We consider the reason for this is that, as shown in the simulation results in Section III, the similarity-based method cannot detect RFI with either H or V polarization alone, while our method can detect RFI regardless of its polarization.

Our DoP-based method can detect RFI both in the time and frequency domains, and the experimental results showed a difference in the performance of the two domains. Actual observation data contains RFI signals with a long duration, and suppressing them with our method in the time domain sometimes causes excessive signal loss of the radar echoes. Our method in the time domain strongly suppresses signals,

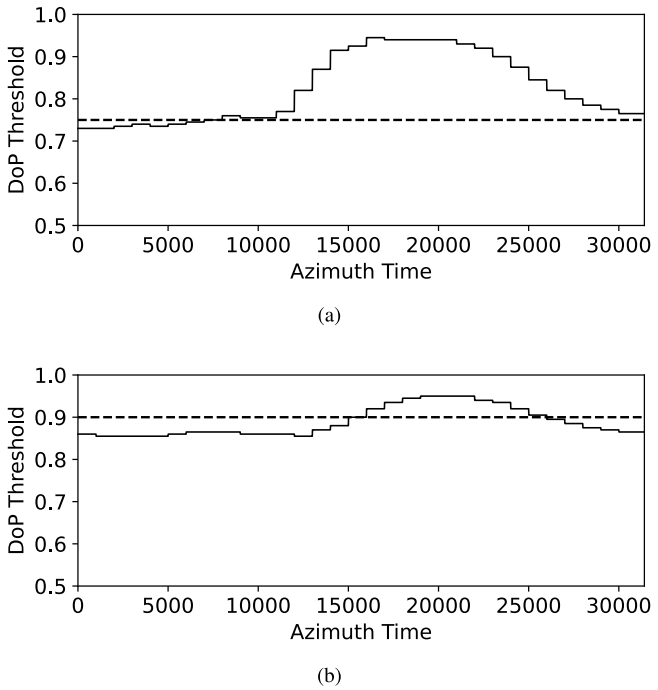


Fig. 20. Actual threshold values of the adaptive DoP method for ALPSRP071951440 (HH and HV). (a) Time domain. (b) Frequency domain. Dashed lines indicate the fixed threshold values at 0.75 for the time domain and 0.9 for the frequency domain.

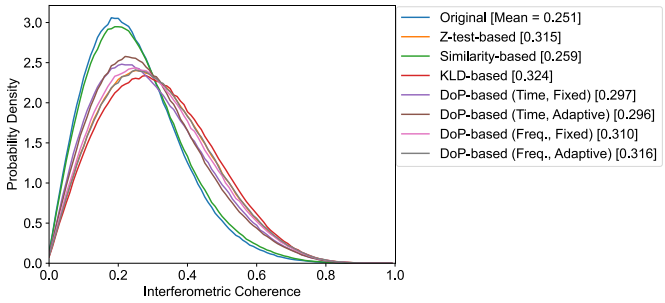


Fig. 21. Histograms and mean values of the interferometric coherence between cross-polarizations (HV, VH) SLC images for ALPSRP071951440.

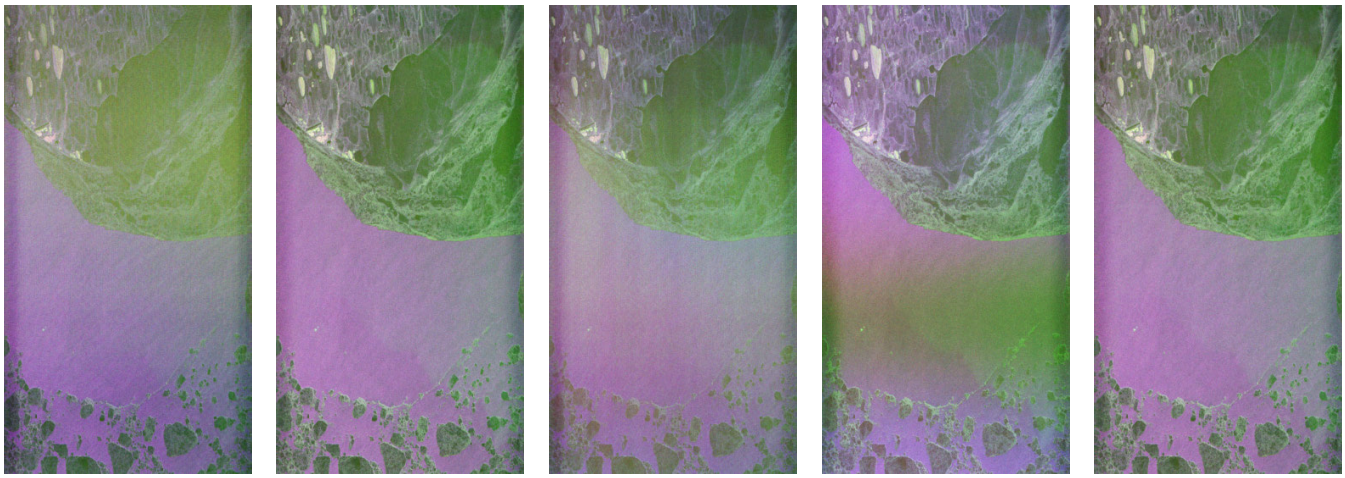
so our method in the frequency domain is more suitable for mitigating long-duration or narrowband RFI.

The differences in the performance of our DoP-based method in the time and frequency domains were more significant in the case of ALPSRP071951440 than in ALPSRP172601440. In scene ALPSRP071951440, the DoP-based method in the frequency domain was more effective than in the time domain. This is due to the large sea area with coherent Bragg scattering in scene ALPSRP071951440. Consistent scatterers spread over a wide area result in high DoP values of the radar echoes. In such cases, our DoP-based method in the time domain may have difficulty in separating RFI from the radar echoes. In contrast, our method in the frequency domain can distinguish RFI from radar echoes with high DoP, as long as the bandwidth of RFI is narrower than that of the radar echoes.

C. Discussion on the Faraday Rotation Effect

For L-band SAR, the Faraday rotation caused by the magnetosphere and ionosphere should be considered when analyzing

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:01:14 UTC from IEEE Xplore. Restrictions apply.



(a) (b) (c) (d) (e)

Fig. 22. Comparison of Pauli decomposition images for ALPSRP071951440. (a) Before RFI mitigation. After RFI mitigation with (b) Z-test-based, (c) similarity-based, (d) KLD-based, and (e) frequency-domain DoP-based method with adaptive threshold method. The area within 10 000th–25 000th azimuth lines is shown in the figures.

the polarization states of received signals. Radar echoes are affected by the Faraday rotation twice during the round-trip propagation. In contrast, RFI does not necessarily undergo the same Faraday rotation as radar echoes because it usually propagates one way either from the ground or the space. Although the Faraday rotation effect may be different for radar echoes and RFI, we assume that the effect is negligible in RFI detection as long as the rotation effect is consistent within each RFI pulse. In short, the Faraday rotation changes the polarization plane, but it does not affect the DoP.

VI. Conclusion

In this article, we proposed an RFI detection method using the DoP, which can be applied in both the time and frequency domains. The simulation and experimental results showed that our method achieved better performance than the conventional methods. We followed the RFI detection approach using two polarizations and demonstrated that the polarization state and DoP are useful for RFI detection. Our DoP-based method exploits the fundamental characteristics of RFI, the polarization state, and enables accurate RFI detection in the time domain, which has been difficult with conventional methods using amplitude or phase information. The time-domain DoP-based method is effective against strong RFI and WBI, and it raises the possibility of on-board RFI detection with its pulse-by-pulse simple computation and fewer sample requirements. On-board RFI detection may be useful for preventing saturation of the receiver and characterizing RFI in real time. In the frequency domain, our method can detect NBI with high sensitivity. It is also effective when the DoP of radar echoes is high, such as in the presence of homogeneous scatterers. We also proposed an adaptive thresholding method for the DoP, and the experimental results showed that it improved the sensitivity of our method. In future work, investigations will be conducted for other PolSAR satellites such as Sentinel-1, RADARSAT-2, and NASA-ISRO Synthetic Aperture Radar (NISAR). We will analyze the characteristics of actual RFI and perform simulations based on them to further improve the performance of our method.

Acknowledgment

The authors acknowledge the stimulated discussion in the meeting of the Cooperative Research Project R05/B01 of the Research Institute of Electrical Communication, Tohoku University.

References

[1] Y. Hashimoto, R. Natsuaki, and A. Hirose, "RFI detection using degree of polarization for polarimetric synthetic aperture radar," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2024, pp. 11482–11485.

[2] M. Tao, J. Su, Y. Huang, and L. Wang, "Mitigation of radio frequency interference in synthetic aperture radar data: Current status and future trends," *Remote Sens.*, vol. 11, no. 20, p. 2438, Oct. 2019.

[3] F. van Leijen, B. Rommen, M. Davidson, and R. Hanssen, "The impact of ground-based uncorrelated radio frequency interference (RFI) sources on satellite radar interferometric ground motion analysis," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2015, pp. 4093–4096.

[4] M. Tao, F. Zhou, and Z. Zhang, "Characterization and mitigation of radio frequency interference in PolSAR data," *Radio Sci.*, vol. 52, no. 11, pp. 1405–1418, Nov. 2017. [Online]. Available: https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1002/2017RS006252

[5] T. Miller, L. Potter, and J. McCorkle, "RFI suppression for ultra wideband radar," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 33, no. 4, pp. 1142–1156, Oct. 1997.

[6] A. Potsis, A. Reigber, and K. P. Papathanassiou, "A phase preserving method for RF interference suppression in P-band synthetic aperture radar interferometric data," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Mar. 1999, pp. 2655–2657.

[7] Z. Liu, G. Liao, and Z. Yang, "Time variant RFI suppression for SAR using iterative adaptive approach," *IEEE Geosci. Remote Sens. Lett.*, vol. 10, no. 6, pp. 1424–1428, Nov. 2013.

[8] S. Joy, L. H. Nguyen, and T. D. Tran, "Radio frequency interference suppression in ultra-wideband synthetic aperture radar using range-azimuth sparse and low-rank model," in *Proc. IEEE Radar Conf. (RadarConf)*, Philadelphia, PA, USA, May 2016, pp. 1–4.

[9] H. Yang, C. Chen, S. Chen, F. Xi, and Z. Liu, "A dictionary-based SAR RFI suppression method via robust PCA and chirp scaling algorithm," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 7, pp. 1229–1233, Jul. 2021.

[10] H. Liu, D. Li, Y. Zhou, and T.-K. Truong, "Simultaneous radio frequency and wideband interference suppression in SAR signals via sparsity exploitation in time-frequency domain," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 5780–5793, Oct. 2018.

[11] L. H. Nguyen and T. D. Tran, "Efficient and robust RFI extraction via sparse recovery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 6, pp. 2104–2117, Jun. 2016.

[12] J. Ren, T. Zhang, J. Li, L. H. Nguyen, and P. Stoica, "RFI mitigation for UWB radar via hyperparameter-free sparse SPICE methods," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 6, pp. 3105–3118, Jun. 2019.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:01:14 UTC from IEEE Xplore. Restrictions apply.

[13] Y. Huang, Z. Chen, C. Wen, J. Li, X.-G. Xia, and W. Hong, "An efficient radio frequency interference mitigation algorithm in real synthetic aperture radar data," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, doi: 10.1109/TGRS.2022.3155068.

[14] X. Lu et al., "Radio frequency interference suppression for SAR via block sparse Bayesian learning," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 12, pp. 4835–4847, Dec. 2018.

[15] S. Buckreuss and R. Horn, "E-SAR P-band SAR subsystem design and RF-interference suppression," in *Proc. Sens. Manag. Environ. IEEE Int. Geosci. Remote Sensing. Symp. (IGARSS)*, vol. 1, Jul. 1998, pp. 466–468.

[16] P. A. Rosen, S. Hensley, and C. Le, "Observations and mitigation of RFI in ALOS PALSAR SAR data: Implications for the DESDynI mission," in *Proc. IEEE Radar Conf.*, May 2008, pp. 1–6.

[17] M. Shimada, O. Isoguchi, T. Tadono, and K. Isono, "PALSAR radiometric and geometric calibration," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 12, pp. 3915–3932, Dec. 2009.

[18] J. Feng, H. Zheng, Y. Deng, and D. Gao, "Application of subband spectral cancellation for SAR narrow-band interference suppression," *IEEE Geosci. Remote Sens. Lett.*, vol. 9, no. 2, pp. 190–193, Mar. 2012.

[19] F. J. Meyer, J. B. Nicoll, and A. P. Doulgeris, "Correction and characterization of radio frequency interference signatures in L-band synthetic aperture radar data," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 10, pp. 4961–4972, Oct. 2013.

[20] A. Monti-Guarnieri, D. Giudici, and A. Recchia, "Identification of C-band radio frequency interferences from Sentinel-1 data," *Remote Sens.*, vol. 9, no. 11, p. 1183, Nov. 2017.

[21] R. Natsuaki, T. Motohka, M. Watanabe, M. Shimada, and S. Suzuki, "An autocorrelation-based radio frequency interference detection and removal method in azimuth-frequency domain for SAR image," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 12, pp. 5736–5751, Dec. 2017.

[22] F. Zhou, R. Wu, M. Xing, and Z. Bao, "Eigensubspace-based filtering with application in narrow-band interference suppression for SAR," *IEEE Geosci. Remote Sens. Lett.*, vol. 4, no. 1, pp. 75–79, Jan. 2007.

[23] N. Li, Z. Lv, and Z. Guo, "Pulse RFI mitigation in synthetic aperture radar data via a three-step approach: Location, notch, and recovery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, doi: 10.1109/TGRS.2022.3161368.

[24] F. Zhou, M. Xing, X. Bai, G. Sun, and Z. Bao, "Narrow-band interference suppression for SAR based on complex empirical mode decomposition," *IEEE Geosci. Remote Sens. Lett.*, vol. 6, no. 3, pp. 423–427, Jul. 2009.

[25] V. T. Vu, T. K. Sjögren, M. I. Pettersson, L. Håkansson, A. Gustavsson, and L. M. H. Ulander, "RFI suppression in ultrawideband SAR using an adaptive line enhancer," *IEEE Geosci. Remote Sens. Lett.*, vol. 7, no. 4, pp. 694–698, Oct. 2010.

[26] T. Bollian, B. Osmanoglu, R. Rincon, S.-K. Lee, and T. Fatoyinbo, "Adaptive antenna pattern notching of interference in synthetic aperture radar data using digital beamforming," *Remote Sens.*, vol. 11, no. 11, p. 1346, Jun. 2019. [Online]. Available: https://www.mdpi.com/2072-4292/11/11/1346

[27] T. Bollian and M. Younis, "On-board RFI detection for reflector-based multichannel SAR systems," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2021, pp. 1670–1672.

[28] R. Natsuaki and A. Hirose, "Polarimetric analysis of RFI in L-band SAR system," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Kuala Lumpur, Malaysia, Jul. 2022, pp. 5145–5148.

[29] J. Wang, L. Zhang, Z. Meng, S. Wei, and L.-L. Chen, "General RFI suppression with sidelobe cancellation filtering for dual-polarization SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[30] A. Uozumi, A. Hirose, and R. Natsuaki, "RFI detection using cross-polarization coherence for dual-and full-polarimetric synthetic aperture radar," in *Proc. 35th Gen. Assem. Scientific Symp. Int. Union Radio Sci. (URSI GASS)*, Aug. 2023, pp. 1–4.

[31] Y. Mao et al., "Radio frequency interference mitigation in SAR systems via multi-polarization framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, doi: 10.1109/TGRS.2024.3384953.

[32] C. Guo, X. Wu, C. Feng, and Z. Zeng, "Spectrum sensing for cognitive radios based on directional statistics of polarization vectors," *IEEE J. Sel. Areas Commun.*, vol. 31, no. 3, pp. 379–393, Mar. 2013.

[33] S. Yatawatta, "Polarization-based online interference mitigation in radio interferometry," in *Proc. 28th Eur. Signal Process. Conf. (EUSIPCO)*, Jan. 2021, pp. 1961–1965.



Yu Hashimoto (Graduate Student Member, IEEE) received the B.S. degree in electrical engineering from The University of Tokyo, Tokyo, Japan, in 2024, where he is currently pursuing the M.S. degree in electrical engineering and information systems.

His research interests include synthetic aperture radar, radio frequency interference detection and mitigation, and signal processing.



Akira Hirose (Fellow, IEEE) received the Ph.D. degree in electronic engineering from The University of Tokyo, Tokyo, Japan, in 1991.

In 1987, he joined the Research Center for Advanced Science and Technology (RCAST), The University of Tokyo, as a Research Associate, where he was appointed as an Instructor in 1991. From 1993 to 1995, on leave from The University of Tokyo, he joined the Institute for Neuroinformatics, University of Bonn, Bonn, Germany. He is currently a Professor with the Department of Electrical Engineering and Information Systems, The University of Tokyo. He authored or co-authored several books, such as *Complex-Valued Neural Networks* (New York, NY, USA: Springer, 2012, second edition). His research interests include wireless electronics and neural networks.

Dr. Hirose is a fellow of the Institute of Electronics, Information and Communication Engineers (IEICE) and a member of the Japanese Neural Network Society (JNNS) and the Asia-Pacific Neural Network Society (APNNS). Since 2009, he has been a member of the IEEE Computational Intelligence Society (CIS) Neural Networks Technical Committee (NNTC). Since 2006, he has been a Governing Board Member of the Asia-Pacific Neural Network Assembly (APNNA), which was transformed to APNNS in 2016. In 2016, he was the Founding President of the APNNS. From 2013 to 2015, he was the President of the JNNS. From 2013 to 2015, he was the Vice President of the IEICE Electronics Society. From 2013 to 2015, he was the IEEE Geoscience and Remote Sensing Society's All Japan Chapter Chair. From 2017 to 2018, he was also the IEEE CIS All Japan Chapter Chair. He was the General Chair for the 2013 Asia-Pacific Conference on Synthetic Aperture Radar, 2016 International Conference on Neural Information Processing, 2019 International Geoscience and Remote Sensing Symposium, and 2024 IEEE World Congress on Computational Intelligence. Since 2010, he has been the Founding Chair for the NNTC Complex-Valued Neural Network Task Force. Since 2024, he has been the Editor-in-Chief (EiC) of the *IEICE Transactions on Electronics* and *IEICE Electronics Express* (ELEX). He was an Associate Editor of IEEE TRANSACTIONS ON NEURAL NETWORKS from 2009 to 2011, IEEE GEOSCIENCE AND REMOTE SENSING NEWSLETTER from 2009 to 2012, and IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS from 2019 to 2022. He has been a Senior Editor of IEEE ACCESS since 2024.



Ryo Natsuaki (Senior Member, IEEE) received the B.S., M.S., and Ph.D. degrees in electrical engineering from The University of Tokyo, Tokyo, Japan, in 2009, 2011, and 2014, respectively.

He was an Aerospace Project Research Associate of Japan Aerospace Exploration Agency (JAXA), Tsukuba, Japan, from 2014 to 2017. He was also a Guest Scientist with the Microwaves and Radar Institute, German Aerospace Center (DLR), Oberpfaffenhofen, Germany, from 2018 to 2020, under the JSPS Overseas Research Fellowships. He is currently an Associate Professor with the Department of Electrical Engineering and Information Systems, The University of Tokyo. His research interests include active remote sensing with synthetic aperture radar (SAR).

Dr. Natsuaki is a member of the Institute of Electronics, Information and Communication Engineers (IEICE). He is also a Technical Committee Member of Frequency Allocations in Remote Sensing (FARS) and Remote Sensing Environment, Analysis and Climate Technologies Technical Committee (REACT) in GRSS. He was the Publicity Chair for the IEEE International Geoscience and Remote Sensing Symposium (IGARSS) 2019 and the Secretary of the IEEE GRSS Japan Chapter from 2014 to 2017 and the IEICE Technical Committee on Space, Aeronautical and Navigational Electronics (SANE) and Electromagnetic Theory (EMT). He currently serves as an Associate Editor for IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (IEEE TGRS).

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:01:14 UTC from IEEE Xplore. Restrictions apply.